Trace-Class Renewal Dynamics with an Entire Fredholm Determinant

Hénon Route-A Working Series, C142

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Abstract

We construct a countable renewal graph whose natural operator on $\ell^2(\mathbb{N}_0)$ is infinite-rank and trace class. A rank-one factorization gives the all-order identity

$$\det_F(I - zT) = 1 - \sum_{m\geq 1} 2^{-m(m+1)/2} z^m,$$

an entire function of order zero. Every closed path decomposes into first-return excursions, yielding an intrinsic primitive-necklace product. A matched constant-advance control retains a rational formal renewal series but is noncompact, showing that scalar renewal algebra does not by itself establish Fredholm ownership. These are source-dynamical results, not an arithmetic or target spectral identification.

1 A countable source model

Let $\mathcal{H} = \ell^2(\mathbb{N}_0)$ with basis $(e_n)_{n\geq 0}$. Freeze

$$Se_n = b_n e_{n+1}, \qquad Re_n = a_n e_0, \qquad a_n = b_n = 2^{-(n+1)}, \qquad T = S + R.$$

The directed graph has an advance edge $n \to n+1$ of weight b_n and a return edge $n \to 0$ of weight a_n. One edge is one unit of clock. The unique first-return excursion of length m has weight

$$c_m = a_{m-1} \prod_{j=0}^{m-2} b_j = 2^{-m(m+1)/2}. \tag{1}$$

No parameter is fitted and no external spectral data enter this definition.

2 Trace class and the all-order determinant

Theorem 1. *The operator T is trace class, with $\|T\|_1 \leq 1 + 1/\sqrt{3}$, and for all $z \in \mathbb{C}$,*

$$D(z) := \det_F(I - zT) = 1 - \sum_{m\geq 1} c_m z^m. \tag{2}$$

The function D is entire of order zero.

Proof. The singular values of S are (b_n), so $\|S\|_1 = 1$. Writing $R = |e_0\rangle\langle a|$ gives $\|R\|_1 = \|a\|_2 = 1/\sqrt{3}$. The shift S is quasinilpotent and every S^k has zero trace, hence $\det_F(I - zS) = 1$. Factoring through $I - zS$ and applying the rank-one determinant identity gives

$$D(z) = 1 - z\langle a, (I - zS)^{-1} e_0\rangle = 1 - \sum_{m\geq 1} a_{m-1} \prod_{j=0}^{m-2} b_j z^m,$$

which is (2). If $d_m = -2^{-m(m+1)/2}$ denotes its nonconstant coefficient, then the coefficient characterization of entire order gives

$$\limsup_{m\to\infty} \frac{m\log m}{\log(1/|d_m|)} = \limsup_{m\to\infty} \frac{m\log m}{m(m+1)\log 2/2} = 0.$$

Thus the series converges on all of $\mathbb{C}$ and has order zero. $\square$

The theorem is genuinely infinite-dimensional. If T_N denotes the compression to $e_0, \dots, e_{N-1}$, the same factorization yields

$$\det(I - zT_N) = 1 - \sum_{m=1}^{N} c_m z^m, \qquad \operatorname{Tr}(T_N^n) = \operatorname{Tr}(T^n) \quad (n \le N). \tag{3}$$

A closed path of length n cannot reach vertex n; this proves the trace stability rather than merely observing it numerically.

3 Primitive excursion product

Every closed path must hit vertex zero, because advances alone never close. Cutting at consecutive visits to zero gives a cyclic word $p = (m_1, \dots, m_k)$ in positive excursion lengths. Put $\ell_p = \sum_j m_j$ and $w_p = \prod_j c_{m_j}$.

Proposition 1. *On $|z| < (1 + 1/\sqrt{3})^{-1}$,*

$$D(z) = \prod_{[p]\ \text{primitive}} (1 - w_p z^{\ell_p}), \tag{4}$$

where $[p]$ is a cyclic necklace. Orientation, total edge clock, weight, and repetition remain explicit.

Proof. The trace-class logarithm $-\log D(z) = \sum_{n\ge1} \operatorname{Tr}(T^n) z^n/n$ is absolutely convergent on the stated disk, since $|\operatorname{Tr}(T^n)| \le \|T^n\|_1 \le \|T\|_1^n$. All graph weights are nonnegative, so the same bound gives absolute convergence before regrouping. The unique excursion decomposition reorganizes rooted closed paths into primitive necklaces and repetitions, giving $\sum_{[p]} \sum_{r\ge1} w_p^r z^{r\ell_p}/r$, the negative logarithm of (4). $\square$

4 An ownership control

Keep $a_n = 2^{-(n+1)}$ but replace every advance weight by $\widetilde{b}_n = 1/2$. The formal first-return coefficients become $2^{-(2m-1)}$, so formal summation gives

$$1 - \sum_{m\ge1} 2^{-(2m-1)} z^m = \frac{1 - 3z/4}{1 - z/4}.$$

Nevertheless, the associated shift has singular value $1/2$ with infinite multiplicity and is noncompact. A rank-one return perturbation cannot make it compact. Thus this rational expression is not the ordinary Fredholm determinant of the natural control operator. The distinction is operator ownership, not algebraic complexity.

5 Exact audit and Route-A boundary

The release reconstructs 16 coefficients, 12 traces, and all 225 primitive necklaces through total clock ten. A producer-independent checker makes 110 assertions; a separate SymPy run makes 56 exact checks; canonical replay is byte-identical; and 24 repaired-hash plus one stale-hash mutation are rejected. Finite ledgers are implementation sentinels, not proofs of the infinite statements.

The conservative verdict is

$$(\mathrm{A1\_WEAK}, \mathrm{A2\_FAIL}, \mathrm{A3\_FAIL}, \mathrm{A4\_FAIL}).$$

The source determinant is entire, but no target divisor, target functional equation, counting law, prime correspondence, arithmetic local factor, Euler factor, root number, automorphy object, unitary lift, or Hilbert–Pólya operator is supplied. Route B remains unauthorized under `NO_BAD_EULER_OR_ROOT_NUMBER`.

6 Conclusion

C142 supplies a countable, infinite-rank dynamical owner whose trace law, Fredholm determinant, and primitive cycles are one coherent object. Any target-facing comparison is a separate, currently unsupported problem.